Bernoulli NML Selects Relative-Periodic Domains

A fitted relative-periodic scaffold isolates defect masks that retain visible structure in several 2D cellular automata

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Affiliation to be added

Main result. A single Bernoulli-NML selector fits a global relative-periodic scaffold to finite spacetimes in 1D, 2D, and 3D cellular automata. In the strongest 2D cases, the remaining defect mask is sparse, but it does not collapse to featureless noise.

Problem

Cellular automata often contain broad domains together with particles, interfaces, and other persistent defects. For a finite spacetime, the practical question is which global period and shift give the best explanation of the observed pattern.

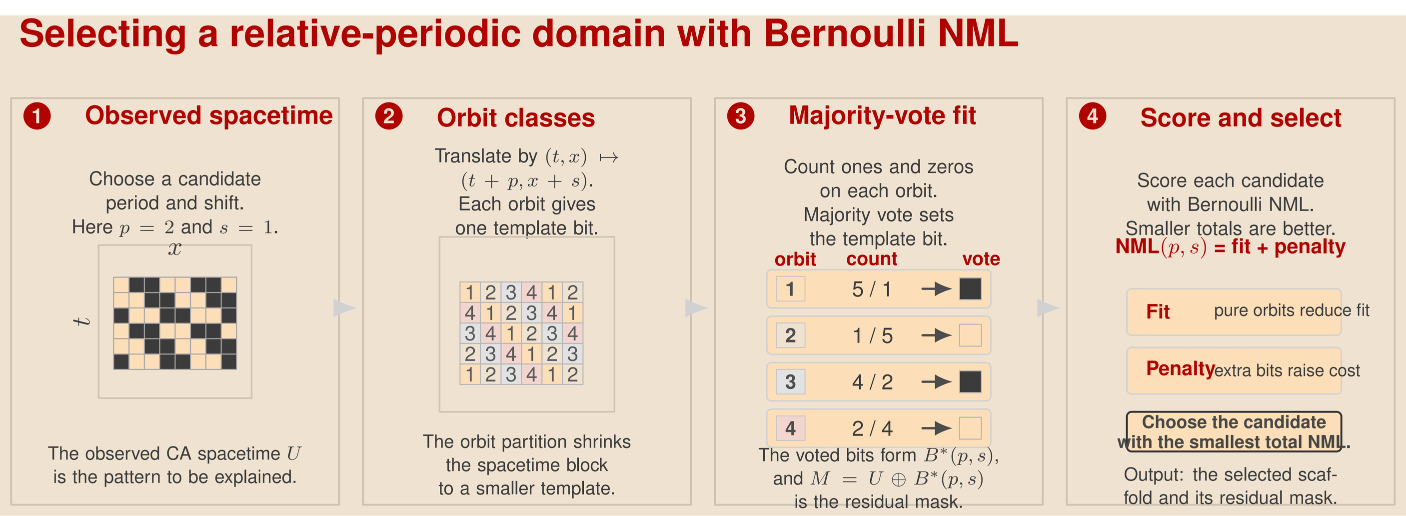
Guiding idea. First fit the global scaffold, and then inspect the residual mask that remains.

Selector

- For a candidate period p and shift $\mathbf{s}$, the translation $(t, \mathbf{x}) \mapsto (t + p, \mathbf{x} + \mathbf{s})$ partitions cells into orbit classes.
- Majority vote on each orbit yields a fitted background, denoted $B^*(p, \mathbf{s})$.
- If U denotes the observed spacetime, then the defect mask is $M = U \oplus B^*(p, \mathbf{s})$.
- Bernoulli NML ranks candidate periods and shifts while penalizing unnecessary template size.

$$\mathrm{NML}(p, \mathbf{s}) = \sum_j n_j H_b(\hat{\theta}_j) + \frac{1}{2}\sum_j \log_2 n_j$$

Here n_j is the size of orbit class j, and $\hat{\theta}_j$ is its fitted Bernoulli parameter. Pure orbit classes lower the fit term, but extra template bits raise the penalty.



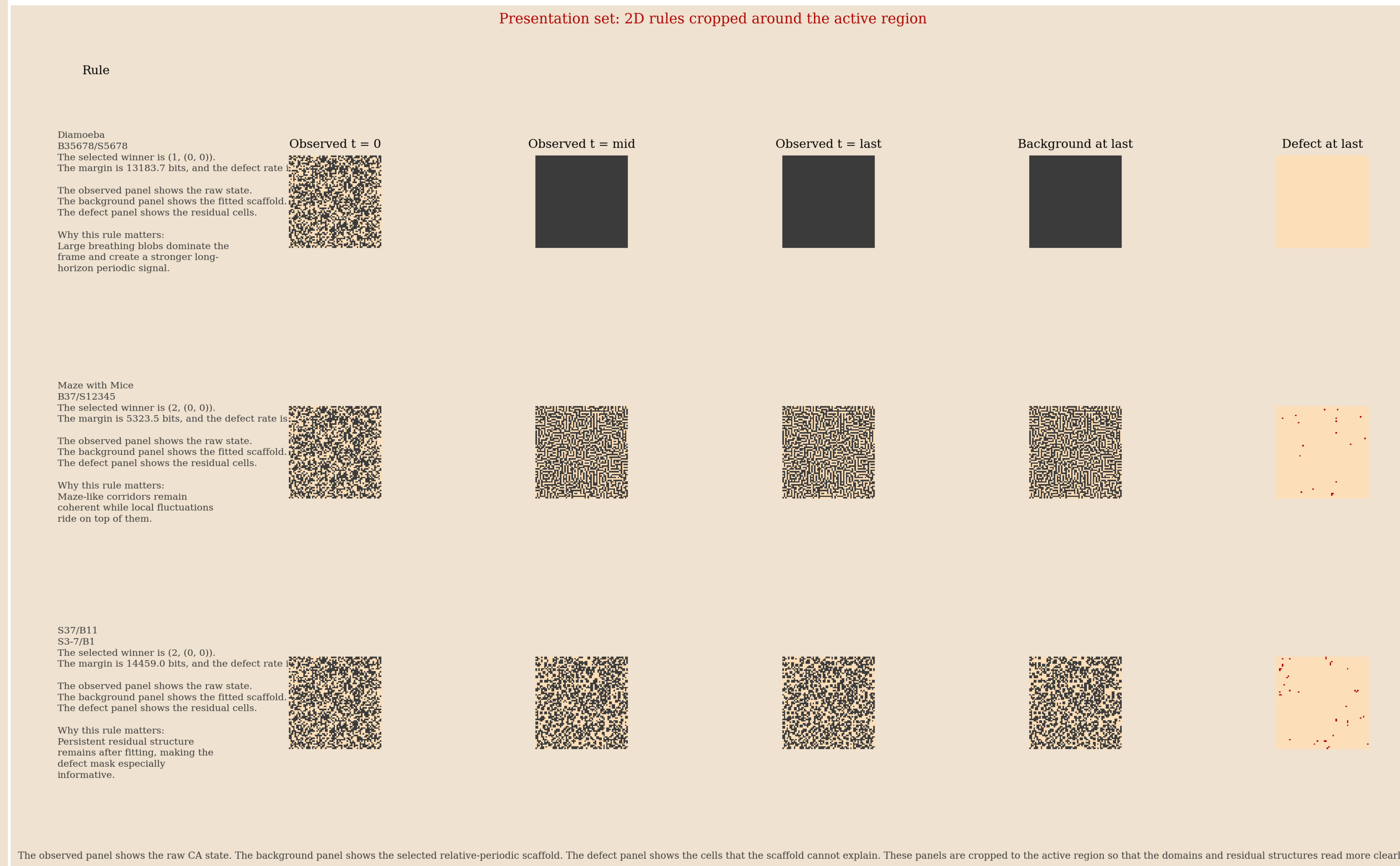
Why Raw Likelihood Fails

Selector	Outcome on 105 nontrivial LifeWiki rules at $T = 100$
Residual minimization	16 pick $p = 1$, 8 pick $p = 2$, and 81 pick period ≥ 4
Bernoulli NLL	All 105 pick the maximum tested period
Bernoulli NML	97 pick $p = 1$, 7 pick $p = 2$, and 1 picks $p = 8$

Raw Bernoulli likelihood rewards large templates because they can memorize the data. The NML penalty removes that failure mode.

2D Main Result

These panels distinguish the broad scaffold from the residual structure that remains after subtraction.



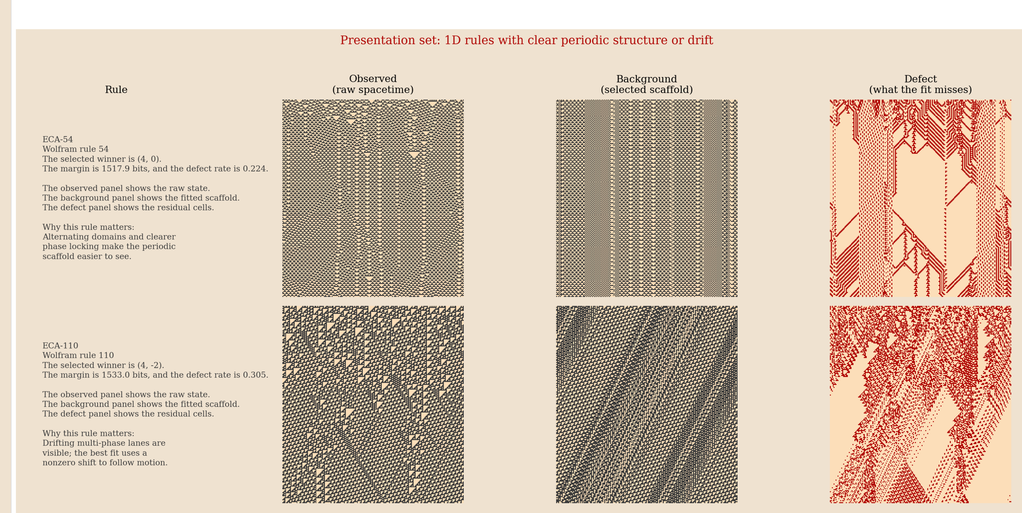
The observed panel shows the raw CA state. The background panel shows the selected relative-periodic scaffold. The defect panel shows the cells that the scaffold cannot explain. These panels are cropped to the active region so that the domains and residual structures read more clearly.

Main visual claim. In 2D, the fitted scaffold explains broad regular domains, but the residual mask does not become featureless. Instead, it often retains sparse coherent structure.

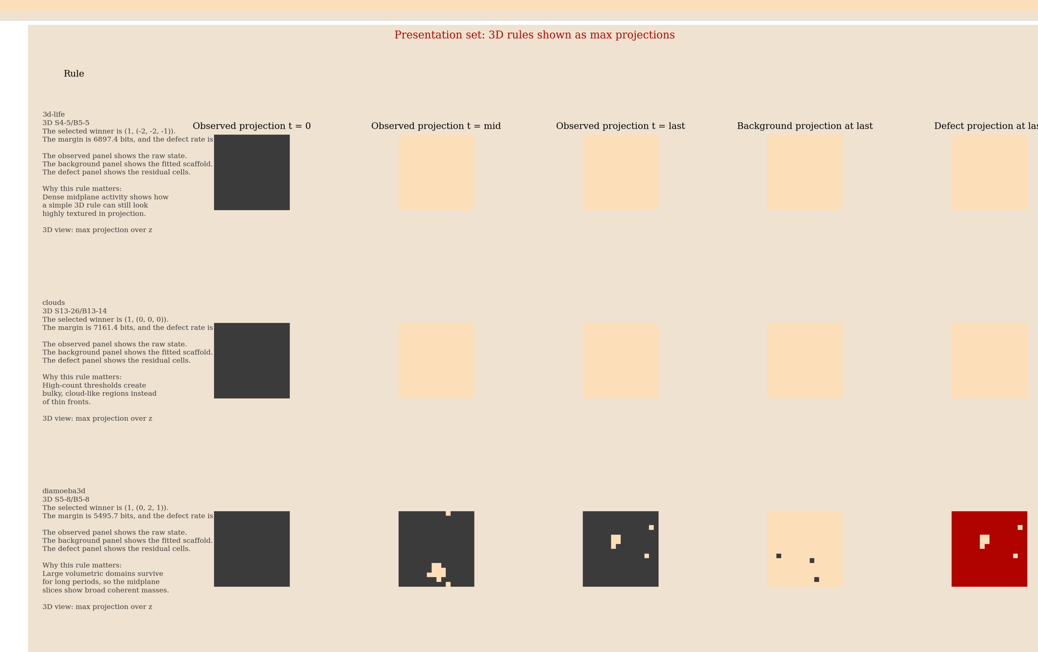
Focal 2D Cases

- In **Diamoeba**, large breathing domains lead the selector to choose $(p, \mathbf{s}) = (1, (0, 0))$, with defect rate 0.077.
- In **Maze with Mice**, a coherent maze background yields winner $(2, (0, 0))$, with defect rate 0.008.
- In **S37/B11**, a low-defect background coexists with persistent residual structure; the presentation run gives winner $(2, (0, 0))$ and defect rate 0.015.

Cross-Dimensional Checks



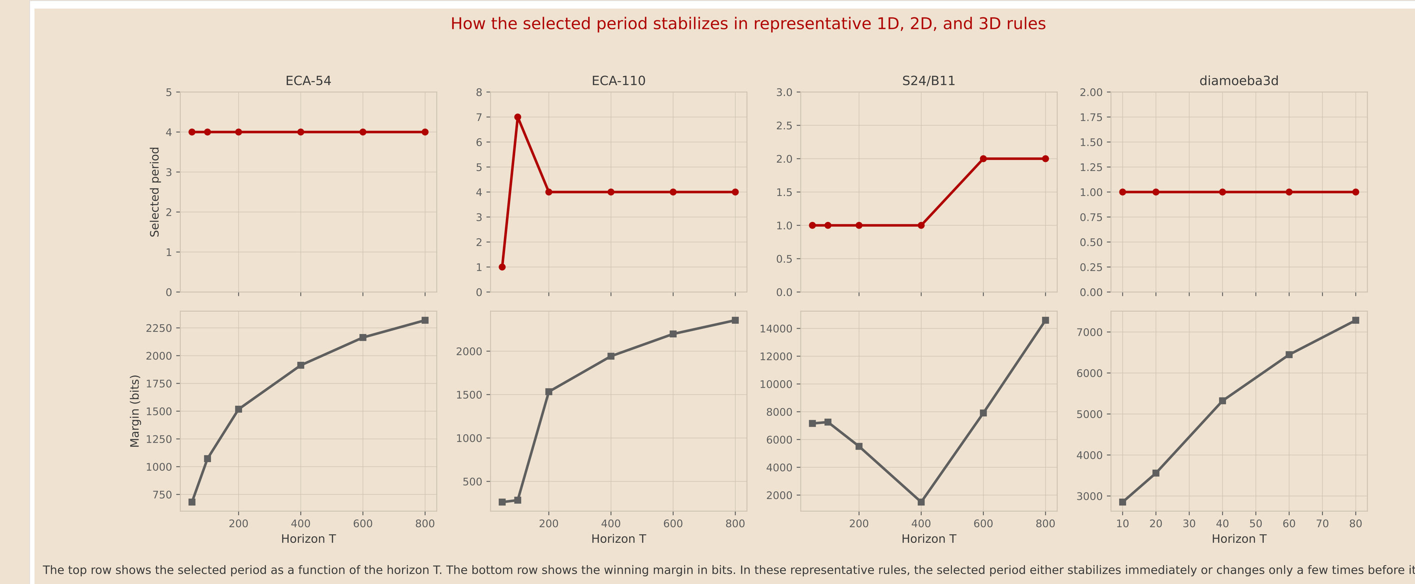
1D. ECA-54 locks to period 4. Under the period-first selector, ECA-110 stabilizes at $(p, s) = (4, -2)$.



3D. The same decomposition still works for volumetric rules; the tested diamoeba3d run settles to period 1.

Stability Across Horizon

As the observation horizon grows, the selected period usually stabilizes quickly.



The top row shows the selected period as a function of the horizon T. The bottom row shows the winning margin in bits. In these representative rules, the selected period either stabilizes immediately or changes only a few times before it locks.

- **ECA-54** locks immediately to period 4.
- **ECA-110** passes through a short transient and then locks to period 4 with shift -2.
- **S24/B11** changes later, because longer horizons reveal additional periodic structure.
- **Diamoeba3d** corrects a small-sample transient and then locks to period 1.

Controls and Robustness

2D summary	Mean period	Mean defect rate
Original runs	2.2	0.014
Bernoulli i.i.d.	1.0	0.384
Space shuffled	1.0	0.384
Time shuffled	1.0	0.114

Shuffled and i.i.d. controls collapse toward period 1 and produce much larger defect rates. That behavior suggests that the selector is responding to genuine spacetime structure rather than to generic occupancy statistics.

Rule	Modal p	Defect rate	Seed consistency
S14/B11	1	0.0059	10 / 10
S24/B11	2	0.0278	10 / 10
S37/B11	2	0.0441	10 / 10
S11/B37	4	0.0275	7 / 10

For **S37/B11**, the late defect density stays near 0.011 from 32^2 to 192^2 grids, while the mean number of late defects grows from 11.4 to 428.5.

Takeaways

- One Bernoulli-NML selector provides usable periodic scaffolds in 1D, 2D, and 3D cellular automata.
- The complexity penalty changes the selected period qualitatively; it is not a cosmetic adjustment.
- In 2D, residual masks expose structured defects and therefore make broader surveys feasible.
- The selector is a front end for richer local analysis, not a replacement for computational mechanics.

Repository and figure sources. The selector diagram is generated from `poster/figures/algorithm_detailed_tikz.tex`. The stabilization figure is generated by `scripts/alife_stabilization_summary.py`. The presentation panels are generated in `outputs/alife_2026/rule_diagrams`.

Supporting data. Control summaries are in `outputs/alife_2026/null_controls`. Multi-seed and scaling summaries are in `outputs/strengthening_v2`.

Project link. `https://github.com/zitterbewegung/relative_symmetry_repair`

Scan for repo, paper assets, and scripts.

Replace with a preprint URL if needed.